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Floating solar farm — Feasibility study

DTU Risø, Roskilde Fjord · Sunlit Sea technology

Prepared: 2026-11-04 · Prepared by: Sunlit Sea AS · Status: Preliminary estimate

Irradiation data: PVGIS 5.2, typical meteorological year

Headline figures

Panel count	Installed capacity	POA irradiation (tilted plane)
47 panels	**0.035 MWp DC**	**1 063 kWh/m²/year**

Performance Ratio	Estimated production	Specific yield
90.0 %	**34.2 MWh/year**	**976 kWh/kWp/year**

1 Panel configuration and area

Installed capacity is provided directly as 35 kWp DC. Panel configuration and area derivation are not performed (no `--area` supplied).

Parameter	Value	Unit	Note
Installed DC capacity	35	kWp	Supplied directly
Assumed panel power	740	Wp	Default
Panel count (derived)	~47	panels	35 × 1 000 ÷ 740

2 Solar irradiation and POA analysis

At 2° south-facing tilt the panels capture slightly more irradiation than a horizontal plane. POA is taken from PVGIS 5.2 (Perez model).

Parameter	Value	Unit	Note
GHI (horizontal reference)	1 045	kWh/m²/year	PVGIS MRcalc, 2005–2020
POA at 2° (tilted plane)	1 063	kWh/m²/year	PVGIS PVcalc
Tilt angle	2	°	Facing south
Gain vs horizontal plane	+1.8	%	POA ÷ GHI – 1

3 Loss budget — PR decomposition

Loss component	Loss (%)	Note
Temperature coefficient	3.0	
Inverter efficiency	2.0	
DC cable loss	1.0	
AC cable loss	0.5	
Mismatch	0.5	
Soiling	0.5	
LID (light-induced degradation)	0.5	
Availability / downtime	1.0	
Transformer loss	1.0	
Total loss	**10.0**	
Performance Ratio (PR)	**90.0 %**	100 − Σ

PVGIS reference value (crystSi model with internal temperature and spectral derating, no external losses): 34.4 MWh/year, implied PR 92.3 %. The component sum is likely conservative compared with the PVGIS model.

4 Energy production

Parameter	Value	Unit	Calculation
Installed DC capacity (P_peak)	35	kWp	
POA irradiation (H)	1 063	kWh/m²/year	
Performance Ratio (PR)	0.90	—	100 − Σ losses (from loss budget)
Estimated production (E)	**34.2**	MWh/year	P × H × PR, plus 2 % FPV cooling benefit
Specific yield	976	kWh/kWp/year	E ÷ P
Equivalent operating hours	976	h/year	= specific yield numerically
Capacity factor	11.1	%	E ÷ (P × 8 760 h)

```
E = 35 kWp x 1 063 kWh/m2/yr x 0.90 x 1.02 = 34.2 MWh/yr
```

5 Sensitivity analysis

Scenario	GHI deviation	PR deviation	Production (MWh/year)
Low	−5 %	−3 pp	31.4
Base	—	—	**34.2**
High	+5 %	+2 pp	36.7

Production range: 31.4 – 36.7 MWh/year (spread ±8 % around the base value).

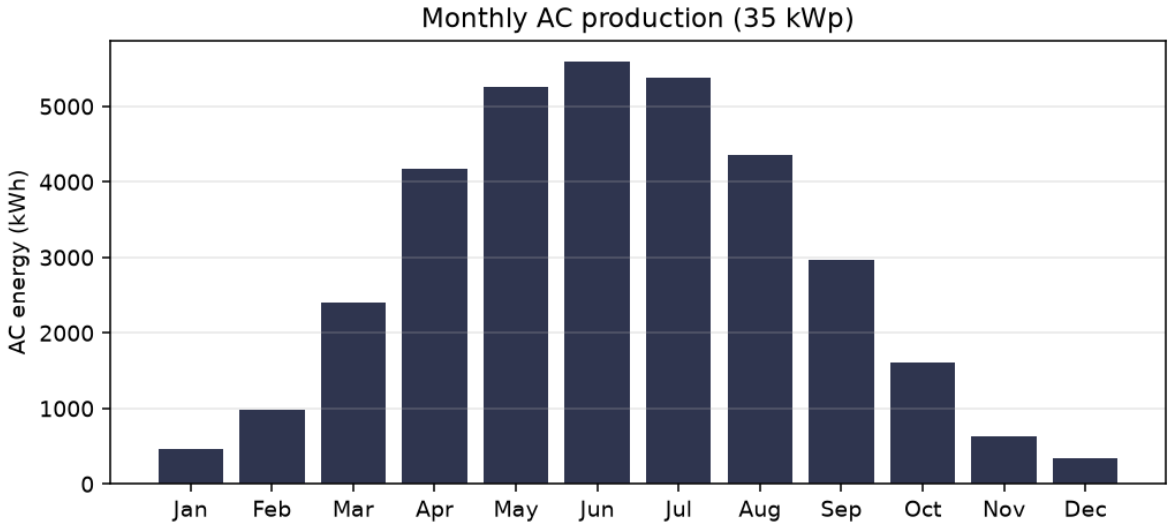


Figure 1: Monthly AC production

6 Key assumptions and limitations

Topic	Assumption / limitation	Recommended next step
Solar irradiation	PVGIS 5.2 (SARAH), regional statistic. No site-specific measurement.	Obtain local pyranometer data from a nearby installation or weather service.
PVGIS lookup point	PVGIS rejected the input coordinates as over-sea; the nearest land grid cell (55.6865, 12.0809) was used. Irradiation data varies little over this distance.	—
POA / tilt	2° tilt assumed. The actual angle is confirmed against the Sunlit Sea design specification at design freeze.	Confirm against the construction drawing.

Topic	Assumption / limitation	Recommended next step
Area	kWp supplied directly; no area dependency.	Fix the net available area after walkways, anchor points and buffer zones.
Panel specification	740 Wp per panel, 2384 × 1303 mm.	Obtain updated product specification at tender time.
Performance Ratio	PR = 90.0 % computed from the component sum. The temperature term may be conservative for floating installations.	Measure the cooling effect at a pilot installation; adjust the FPV cooling bonus parameter.
FPV cooling bonus	2 % DC uplift assumed (literature-based, conservative).	Replace with measurements from a Sunlit Sea pilot installation when available.
Albedo	Water albedo (~6 %) not explicitly modelled; the contribution to flat panels is small.	Can be included as a sensitivity parameter for bifacial panels.
Grid / inverter	AC capacity (MWac) not fixed. DC/AC ratio not defined.	Fix the DC/AC ratio (typical 1.10–1.20 for floating solar).
Regulatory	Regulatory framework for floating solar not assessed in this report.	Initiate regulatory screening for the specific site.

Calculation method: E = P_peak × H × PR × (1 + FPV cooling bonus). Standard method per IEC 61724.

Irradiation basis: POA = 1 063 kWh/m²/year from PVGIS 5.2 (PVGIS 5.2, typical meteorological year), tilted 2° facing azimuth 180° (compass).

This is a preliminary estimate. The results should not be used for investment decisions without further technical due diligence.